

一、选择题(每小题3分,共15分)

1.(B) 2.(D) 3.(B) 4.(C) 5.(B)

二、填空题(每小题3分,共15分)

1. 半径为 R 的上半圆的面积 $\frac{1}{2}\pi R^2$ 2. $0, \frac{100}{2}$ 3. $0, \frac{e^x \ln(1+x^3)}{2}$ 4. $\frac{\pi}{4}$ 5. $\frac{\pi}{4}$

三、解答题(每题7分,共70分)

$$1. \lim_{x \rightarrow 0^+} \frac{\int_0^{x^2} (\tan \sqrt{t} - \sqrt{t}) dt}{\int_0^x t(\tan t - \sin t) dt}$$

$$\begin{aligned} \text{解: } \lim_{x \rightarrow 0^+} \frac{\int_0^{x^2} (\tan \sqrt{t} - \sqrt{t}) dt}{\int_0^x t(\tan t - \sin t) dt} &= \lim_{x \rightarrow 0^+} \frac{2x(\tan x - x)}{x(\tan x - \sin x)} = \lim_{x \rightarrow 0^+} \frac{2(\sec^2 x - 1)}{\sec^2 x - \cos x} \\ &= \lim_{x \rightarrow 0^+} \frac{2 \sin^2 x}{1 - \cos^3 x} = \lim_{x \rightarrow 0^+} \frac{4x}{3 \cos^2 x \sin x} = \frac{4}{3}. \end{aligned}$$

$$2. \text{设 } f(x) = \int_0^x \frac{\sin t}{\pi - t} dt, \text{求 } \int_0^\pi f(x) dx.$$

$$\text{解: } \int_0^\pi f(x) dx = x f(x) \Big|_0^\pi - \int_0^\pi x f'(x) dx = \pi \int_0^\pi \frac{\sin t}{\pi - t} dt - \int_0^\pi x \frac{\sin x}{\pi - x} dx = 2.$$

$$3. \int_0^1 \arctan \sqrt{x} dx$$

$$\text{解: } \int_0^1 \arctan \sqrt{x} dx = x \arctan \sqrt{x} \Big|_0^1 - \int_0^1 \frac{x}{1+x} d\sqrt{x} = \frac{\pi}{4} - (\sqrt{x} - \arctan \sqrt{x}) \Big|_0^1 = \frac{\pi}{2} - 1.$$

$$4. \int_1^2 \frac{1}{x(1+x^4)} dx$$

$$\text{解: } \int_1^2 \frac{1}{x(1+x^4)} dx = \frac{1}{4} \int_1^2 \frac{1}{x^4(1+x^4)} dx^4 = \frac{1}{4} \ln \frac{x^4}{1+x^4} \Big|_1^2 = \frac{5}{4} \ln 2 - \frac{1}{4} \ln 17.$$

$$5. \int_{-1}^1 \frac{2 + \sin x}{1 + x^2} dx$$

$$\text{解: } \int_{-1}^1 \frac{2 + \sin x}{1 + x^2} dx = 4 \int_0^1 \frac{1}{1 + x^2} dx = 4 \arctan x \Big|_0^1 = \pi$$

$$6. \text{设 } f(x) = \begin{cases} e^x, & x \geq 0 \\ 1 + x^2, & x < 0 \end{cases}, \text{求 } \int_{\frac{1}{2}}^2 f(x-1) dx.$$

$$\text{解: 令 } u = x-1, \text{则 } \int_{\frac{1}{2}}^2 f(x-1) dx = \int_{-\frac{1}{2}}^1 f(u) du = \int_{-\frac{1}{2}}^0 (1+x^2) dx + \int_0^1 e^x dx = e - \frac{11}{24}.$$

$$7. \int_0^{+\infty} x e^{-x^2} dx$$

$$\text{解: } \int_0^{+\infty} x e^{-x^2} dx = -\frac{1}{2} \int_0^{+\infty} e^{-x^2} d(-x^2) = -\frac{1}{2} e^{-x^2} \Big|_0^{+\infty} = \frac{1}{2}.$$

8. 设 $f(3x+1) = xe^{\frac{x}{2}}$, 求 $\int_0^1 f(x)dx$.

解: 令 $t = 3x+1$, $f(t) = \frac{t-1}{3}e^{\frac{t-1}{6}}$, $\int_0^1 f(x)dx = \int_0^1 \frac{x-1}{3}e^{\frac{x-1}{6}}dx = 14e^{-\frac{1}{6}} - 12$.

9. $\int_0^{\frac{\pi}{2}} \frac{e^{\sin x}}{e^{\sin x} + e^{\cos x}} dx$.

利用 $\int_0^{\frac{\pi}{2}} f(\sin x)dx = \int_0^{\frac{\pi}{2}} f(\cos x)dx$, 得该积分值为 $\frac{\pi}{4}$.

10. 设 $x_n = \left(\frac{1}{n} \cdot \frac{2}{n} \cdots \frac{n}{n}\right)^{\frac{1}{n}}$, 计算 $\lim_{n \rightarrow \infty} x_n$

解: $\lim_{n \rightarrow \infty} \ln x_n = \lim_{n \rightarrow \infty} \frac{1}{n} (\ln \frac{1}{n} + \ln \frac{2}{n} + \cdots + \ln \frac{n}{n}) = \int_0^1 \ln x dx = x \ln x \Big|_0^1 - \int_0^1 dx = -1$

$$\lim_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} e^{\ln x_n} = e^{-1}.$$

11. 证明不等式 $\ln(1+\sqrt{2}) \leq \int_0^1 \frac{1}{\sqrt{1+x^n}} dx \leq 1$ ($n \geq 2$).

证明: 由于 $\sqrt{1+x^2} \geq \sqrt{1+x^n} \geq 1$, ($0 \leq x \leq 1$)

$$\text{于是 } \int_0^1 \frac{1}{\sqrt{1+x^2}} dx \leq \int_0^1 \frac{1}{\sqrt{1+x^n}} dx \leq \int_0^1 dx$$

$$\text{即 } \ln(1+\sqrt{2}) \leq \int_0^1 \frac{1}{\sqrt{1+x^n}} dx \leq 1.$$

12. 设 $f(x)$ 在 $[0, \pi]$ 上连续, 且 $\int_0^\pi f(x)dx = 0$, $\int_0^\pi f(x) \cos x dx = 0$.

证明: 在 $(0, \pi)$ 内至少存在两个不同的点 ξ_1, ξ_2 , 使 $f(\xi_1) = f(\xi_2)$.

证明:

令 $F(x) = \int_0^x f(t)dt$, 则 $F(0) = F(\pi) = 0$.

$$\text{又 } \int_0^\pi f(x) \cos x dx = \int_0^\pi \cos x dF(x) = F(x) \cos x \Big|_0^\pi + \int_0^\pi F(x) \sin x dx$$

$$= \int_0^\pi F(x) \sin x dx = 0.$$

存在 $\xi \in (0, \pi)$, 使得 $F(\xi) \sin \xi = 0$. 如若不然, 则 $F(x) \sin x$ 在 $(0, \pi)$ 内恒正或恒负, 与已知矛盾.

即有 $F(0) = F(\xi) = F(\pi)$, 所以由罗尔中值定理知, 存在 $\xi_1 \in (0, \xi), \xi_2 \in (\xi, \pi)$,

使得 $F'(\xi_1) = F'(\xi_2)$, 即 $f(\xi_1) = f(\xi_2)$.